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The Physical Direct-Decoration Collar and Compact Multi-Port Response

Research draft for the marked-width MLC program

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Abstract

We resolve an actual physical cell $P \setminus Q_a^\circ$, rather than an ideal equal-footprint rectangle. The physical set splits into an inactive common-ray attachment skeleton and an active Jordan quadrilateral. At an intermediate puzzle boundary, the outgoing and incoming active footprints are generally unequal: only their common atoms are glued, both one-sided differences remain side ports, and the overlap of the inactive attachment arcs remains nonconducting metadata. Compatible Douglas trace spaces give exact fiber-product response composition and functorial last-use elimination. The physical shell resistances satisfy a serial lower bound. Once the finite live port frontier is frozen, the remaining multi-port response has a compact operator limit and an explicit accumulated-resistance estimate. This proves the direct-decoration trace-collar realization and removes an independent analytic end-plumbing gap; the universal horizon, H2, MBTT, UPTCR, and MLC remain open.

1 The physical set and its conducting cell

Let P be the source direct-decoration puzzle piece and let $Q_a \subset P$ be one of its univalent pullbacks. The source Q_a is attached to the two boundary rays of P landing at one vertex. Their common ray portions form a closed arc $A_a \subset \partial P \cap \partial Q_a$, with endpoints x_a^-, x_a^+. Let $y_a^\pm$ be the parent equipotential endpoints and define the parent footprint Γ_a^-, child footprint Γ_a^+, and exposed ray sides $S_a^\pm$ as in Figure 1.

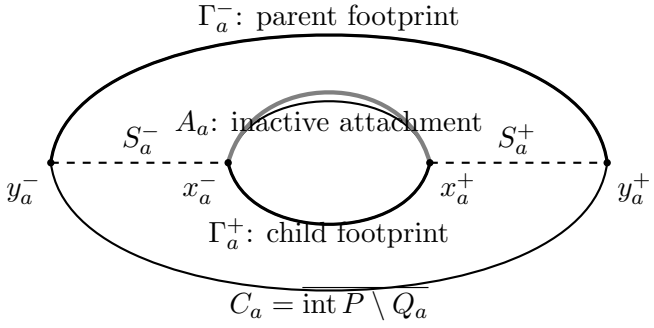


Figure 1: The physical set contains the active quadrilateral C_a and the inactive common-ray attachment A_a. Additional protected or response ports are retained as subarcs of the active boundary.

Put

$$\Omega_a = \operatorname{int} P \setminus Q_a, \qquad C_a = \overline{\Omega_a}.$$

Theorem 1.1 (Active/inactive decomposition)**.** *For nested Jordan disks with common attachment arc A_a,*

$$\boxed{P \setminus Q_a^\circ = C_a \cup A_a, \qquad C_a \cap A_a = \{x_a^-, x_a^+\}.} \tag{1.1}$$

The set C_a is a closed Jordan disk with cyclic boundary

$$\Gamma_a^-, \quad S_a^+, \quad \Gamma_a^+, \quad S_a^-. \tag{1.2}$$

The relative interior of A_a is not approached from Ω_a and therefore carries no independent Dirichlet-to-Neumann trace of the active cell. It is retained as attachment/access metadata.

Proof. At an interior point of a common boundary arc, nested Jordan disks lie on the same side and their interiors agree locally. Hence the conducting complement does not approach the relative interior of A_a. Removing A_a° from ∂P leaves $S_a^- \cup \Gamma_a^- \cup S_a^+$, while removing it from ∂Q_a leaves Γ_a^+. The two residual arcs meet only at $x_a^\pm$ and form the Jordan curve bounding C_a. This also gives (1.1). □

Consequently there is a prime-end conformal map

$$\Psi_a : C_a^\circ \longrightarrow (0, R_a) \times (0, 1)$$

sending the footprints to the vertical sides, where

$$R_a = \mathrm{EL}_{C_a}(\Gamma_a^-, \Gamma_a^+) > 0. \tag{1.3}$$

No Euclidean-length or trace-density equality is asserted. For a word w, $C_{w,a} = \phi_w(C_a)$ has the same modulus and transported marked traces.

If extra boundary contacts occur, overlay the two finite ray/equipotential boundary graphs, subdivide at all vertices, and add finitely many paired crosscuts to make Jordan cells. Exact trace gluing below removes the artificial cuts and recovers the original response.

2 Unequal successive footprints

At an intermediate cylinder $Q = Q_{wa}$, let A^- be the inactive attachment inherited from the preceding cell and A^+ the inactive attachment of the next child. The active footprints are

$$F^{\mathrm{out}} = \overline{\partial Q \setminus A^-}, \qquad F^{\mathrm{in}} = \overline{\partial Q \setminus A^+}. \tag{2.1}$$

Subdivide at all attachment and proof-use endpoints. The relatively open atoms have the exact classification

$$\begin{aligned} I &= \partial Q \setminus (A^- \cup A^+) && \text{common active glue,} \\ E^{\mathrm{out}} &= A^+ \setminus A^- && \text{outgoing-only side port,} \\ E^{\mathrm{in}} &= A^- \setminus A^+ && \text{incoming-only side port,} \\ Z &= A^- \cap A^+ && \text{double-inactive attachment.} \end{aligned} \tag{2.2}$$

Endpoints form a finite fifth class V. Only I is glued; E^{out} and E^{in} remain physical boundary ports; Z is marked but nonconducting.

Freeze a finite proof horizon and insert endpoints of every footprint, intersection, protected collar, obstacle interface, response electrode, ideal interval, and authorized presentation map. The resulting finite trace atoms transport exactly under every inverse word. If the source boundary starts with N_∂ atoms and m marked subarcs are added, there are at most $N_\partial + 2m$ relatively open atoms after duplicate endpoints are removed. Two connected attachment arcs have at most four distinct endpoints, so each ordered symbol pair has at most four open atoms and four endpoint atoms. For an alphabet of size d, the unmarked structural atlas therefore has at most $8d^2$ local types, independently of depth. Any one-sided atom not seeded by a future observable is eliminated at its birth junction by (3.5).

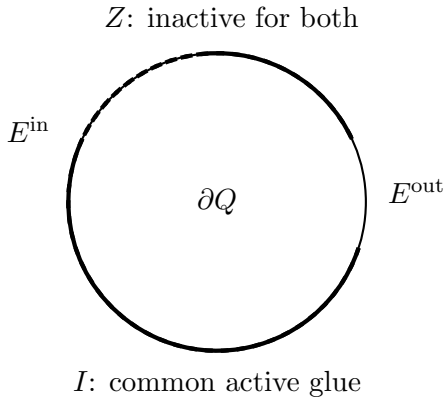


Figure 2: Schematic common refinement. Solid arcs are active from at least one cell; dashed arcs indicate attachment intervals. Actual endpoints are supplied by the puzzle rays and every declared proof-use port.

3 Compatible trace spaces and exact composition

A naive direct sum of $H^{1/2}$ spaces on adjacent arcs can lose endpoint compatibility. For a marked Jordan cell Ω and a finite active boundary union G, use the actual quotient trace space

$$\mathcal{T}_\Omega(G) = \{\operatorname{Tr}_G u : u \in H^1(\Omega)\} \tag{3.1}$$

with its quotient norm. Equivalently, use the Douglas trace space on the prime-end circle and restrict to G. Define

$$q_{\Omega,G}(\tau) = \inf_{\operatorname{Tr}_G u = \tau} \int_\Omega |\nabla u|^2. \tag{3.2}$$

Unlisted active boundary is natural Neumann boundary.

Theorem 3.1 (Fiber-product response composition)**.** *Let two active cells have disjoint interiors and common active trace I. Let r_i be restriction to I and let e_i retain every other live trace label. Then the response of the physical union is*

$$\boxed{q(x_1, x_2) = \inf\{q_1(\tau_1) + q_2(\tau_2) : r_1\tau_1 = r_2\tau_2,\ e_i\tau_i = x_i\}.} \tag{3.3}$$

The one-sided atoms remain among x_1, x_2. Composition is associative and conformally natural.

Proof. Restriction of a finite-energy function gives matching common traces and additive energy. Conversely, two finite-energy functions with matching trace glue to a finite-energy function on the union. The quotient trace spaces retain all endpoint compatibility. Joint infima are independent of elimination order. □

For a continuous forgetful map $p : T \to T_{\text{live}}$, put

$$(p_* q)(y) = \inf_{p\tau = y} q(\tau). \tag{3.4}$$

Theorem 3.2 (Functorial last-use elimination)**.** *For composable forgetful maps,*

$$(r \circ p)_* q = r_*(p_* q). \tag{3.5}$$

Hence a side port may be deleted exactly at its last proof use, even when its trace is endpoint-compatible with neighboring atoms.

If at most b ports are born per stage and every port has future lifetime at most L, the live non-interface frontier has size at most $b(L+1)$.

4 Accumulated physical resistance

For $w = a_1 \cdots a_m$, let

$$C_j = \overline{\operatorname{int} Q_{a_1 \cdots a_{j-1}} \setminus Q_{a_1 \cdots a_j}}, \qquad R_j = R_{a_j}.$$

Every complete outer-to-terminal curve crosses the conducting cells in chronological order.

Theorem 4.1 (Extremal series lower bound)**.** *The selected physical tail resistance satisfies*

$$\boxed{R(w) \geq \sum_{j=1}^{m} R_{a_j}.} \tag{4.1}$$

For a finite root-cell alphabet,

$$R_* := \min_a R_a > 0, \qquad R(w) \geq m R_*. \tag{4.2}$$

Proof. Choose almost extremal metrics ρ_j supported in the disjoint conducting cells, normalized so each cell crossing has length at least one and area close to $1/R_j$. The metric $\sum c_j \rho_j$ gives every complete crossing length at least $\sum c_j$ and area close to $\sum c_j^2/R_j$. Taking $c_j = R_j$ gives (4.1). □

A later wiring that identifies side ports from separated levels may create a bypass and must be included before (4.1) is invoked.

5 Compact multi-port tail response

Freeze the proof-use inventory, eliminate every dead port by (3.5), and choose a stage after the last live non-interface side port. The unequal-footprint prefix is one exact compatible form q_{L} on $X \times \mathcal{H}_{\mathrm{in}}$; let q_{R} be the terminal cap form on $\mathcal{H}_{\mathrm{out}} \times Y$. Compose every remaining physical shell exactly by (3.3). The selected conducting component uniformizes, after any finite overlay cuts are recomposed, to a strip or cylinder of resistance

$$R \geq \sum_{j>N} R_{a_j}. \tag{5.1}$$

Let A be its tangential operator, with kernel the constants and first positive eigenvalue $\lambda_1 > 0$ ($\lambda_1 = \pi$ for the width-one Neumann strip). The complete response is

$$q_R(x,y) = \inf_{\phi,\psi} \left[q_{\mathrm{L}}(x,\phi) + \mathcal{N}_R(\phi,\psi) + q_{\mathrm{R}}(\psi,y) \right]. \tag{5.2}$$

Let $\mathcal{N}_R^{\#}$ retain the exact scalar resistor $1/R$ and the two mean-zero half-cylinder responses. With

$$\eta_A(R) = \frac{2}{e^{\lambda_1 R} - 1},$$

$$(1 - \eta_A(R)) q_R^{\#} \preceq q_R \preceq (1 + \eta_A(R)) q_R^{\#}. \tag{5.3}$$

Define the half-cylinder-closed cap forms

$$\widehat{q}_{\mathrm{L}}(x,c) = \inf_{\xi \perp \mathbf{1}} \left[q_{\mathrm{L}}(x, c\mathbf{1} + \xi) + \langle A\xi, \xi \rangle \right],$$

$$\widehat{q}_{\mathrm{R}}(d,y) = \inf_{\zeta \perp \mathbf{1}} \left[q_{\mathrm{R}}(d\mathbf{1} + \zeta, y) + \langle A\zeta, \zeta \rangle \right].$$

Then exactly

$$q_R^{\#}(x,y) = \inf_{c,d\in\mathbb{C}} \left[\widehat{q}_{\mathrm{L}}(x,c) + \frac{|c-d|^2}{R} + \widehat{q}_{\mathrm{R}}(d,y) \right]. \tag{5.4}$$

Write, modulo the global constant,

$$\widehat{q}_{\mathrm{L}}(x,c) = \langle Lx, x\rangle + 2\Re\langle b_{\mathrm{L}} x, c\rangle + \alpha_{\mathrm{L}}|c|^2$$

and similarly on the right. Positivity gives $\alpha_i = 0 \Rightarrow b_i = 0$.

Theorem 5.1 (Compact multi-port response)**.** *For a fixed physical prefix and terminal cap, or for the finitely many types in a frozen branch graph, q_R extends continuously to $R = \infty$ in normalized response-operator norm. Its parameter image is compact. If $\alpha_{\mathrm{L}}, \alpha_{\mathrm{R}} > 0$, set $\alpha_* = \min(\alpha_{\mathrm{L}}, \alpha_{\mathrm{R}})$. If B bounds the external-to-scalar coupling and M bounds $\|q_R^{\#}\|$, then*

$$\boxed{\|q_R - q_\infty\| \le M\frac{2}{e^{\lambda_1 R} - 1} + \frac{2B^2}{\alpha_*^2 R}.} \tag{5.5}$$

If either scalar cap coefficient is zero, the second term is exactly absent.

Proof. The internal scalar matrix is

$$M_t = \begin{pmatrix} \alpha_{\mathrm{L}} + t & -t \\ -t & \alpha_{\mathrm{R}} + t \end{pmatrix}, \qquad t = R^{-1}.$$

When both coefficients are positive, the resolvent identity gives $\|M_t^{-1} - M_0^{-1}\| \le 2t/\alpha_*^2$, and the scalar Schur complement changes by at most $2B^2/(\alpha_*^2 R)$. If one coefficient vanishes, its coupling vanishes by positivity and direct inversion shows that the visible scalar Schur complement is independent of t. Equation (5.3) contributes at most $2M/(e^{\lambda_1 R} - 1)$. This proves (5.5) and norm continuity. □

For response error at most ε, it is sufficient that

$$\boxed{R \ge \max\left\{ \frac{1}{\lambda_1} \log\left(1 + \frac{4M}{\varepsilon}\right), \frac{4B^2}{\alpha_*^2 \varepsilon} \right\}.} \tag{5.6}$$

Omit the second term in the zero-coefficient cases. A sufficient additional physical depth is the ceiling of the right side divided by R_*.

6 Gate decisions

Theorem 6.1 (Physical direct-decoration trace-collar realization)**.** *Every source direct-decoration $P \setminus Q_a^\circ$ has an exact finite active/inactive ray/equipotential decomposition. Actual unequal footprints, one-sided side ports, double-inactive attachment arcs, protected intervals, obstacles, and response electrodes are retained by the common-refinement atlas; composition and artificial-cut removal are exact; and the complete construction transports under every finite inverse word. Thus*

`DIRECT_DECORATION_TRACE_COLLAR_REALIZATION`

is proved in the source direct-decoration category.

Theorem 6.2 (Corrected end-plumbing trace response)**.** *Once the upstream finite inventory supplies a last live non-interface side-port stage, the physical tail has an exact fixed multi-port prefix, exact full-trace tail response, exact last-use elimination, compact operator limit, and the explicit estimates (5.5)–(5.6). Therefore*

$$\boxed{\textit{END_PLUMBING_TRACE_RESPONSE_CONTROL}}$$

has no remaining independent analytic gap; its remaining dependency is the finite live frontier supplied by `VAAPC-I` *or the horizon theorem.*

For an already supplied direct-decoration branch graph of depth H, the unmarked structural transition atlas has at most $8d^2$ local types and every unseeded one-sided atom is immediately eliminated. With N_0 root trace atoms and alphabet size d, the full raw pullback atlas has at most

$$N_0 \sum_{j=0}^{H} d^j$$

records before dependency closure. This proves the direct-decoration trace portion of frozen input. Global `VAAPC-I` remains open because the source-faithful complete Wave demand graph and final horizon are not yet known.

7 Failure record and remaining path

A failed application must name the first extra contact, inaccessible attachment inserted as a conducting trace, missing refinement endpoint, false glue atom, discarded one-sided side port, electrically misclassified double-inactive atom, phase or endpoint-compatibility defect, side-port bypass, infinite live frontier, zero modulus, failed cap normalization, or insufficient accumulated resistance.

This result does not prove the universal Wave horizon, global `VAAPC-I`, the common primitive, neutral, and virtual response and Value calibration required by H2, the root contraction and discounted error budget required by MBTT, coercivity, rigidity, UPTCR, or MLC. The next response theorem is a universal bounded-lifetime law or future-faithful quotient for the live common-refinement port frontier, followed by uniform comparison with the KKL WAD/Value and neutral sectorial charts.